End-to-End Data Analytics in Power BI: From Data Cleaning to DAX Insights

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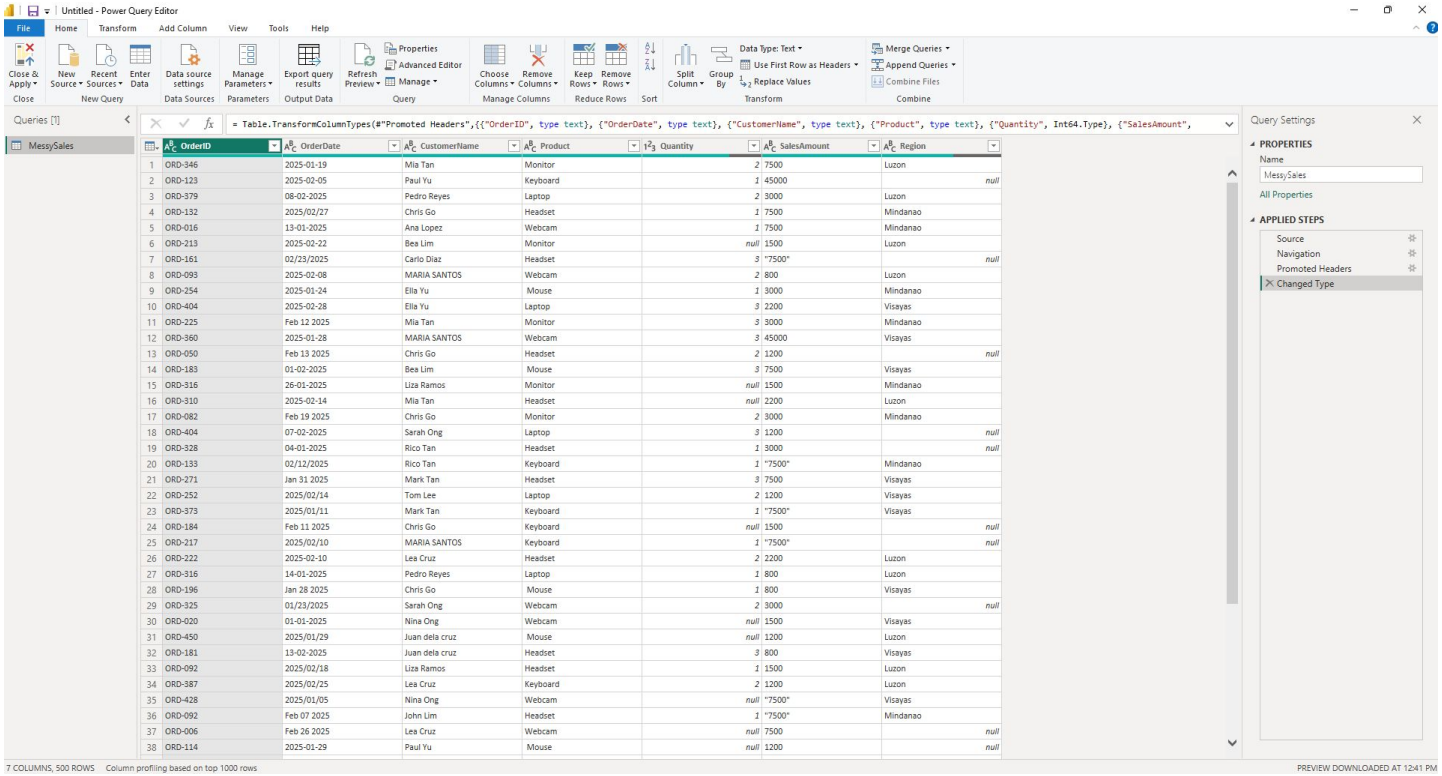
IS 107 - Business Intelligence

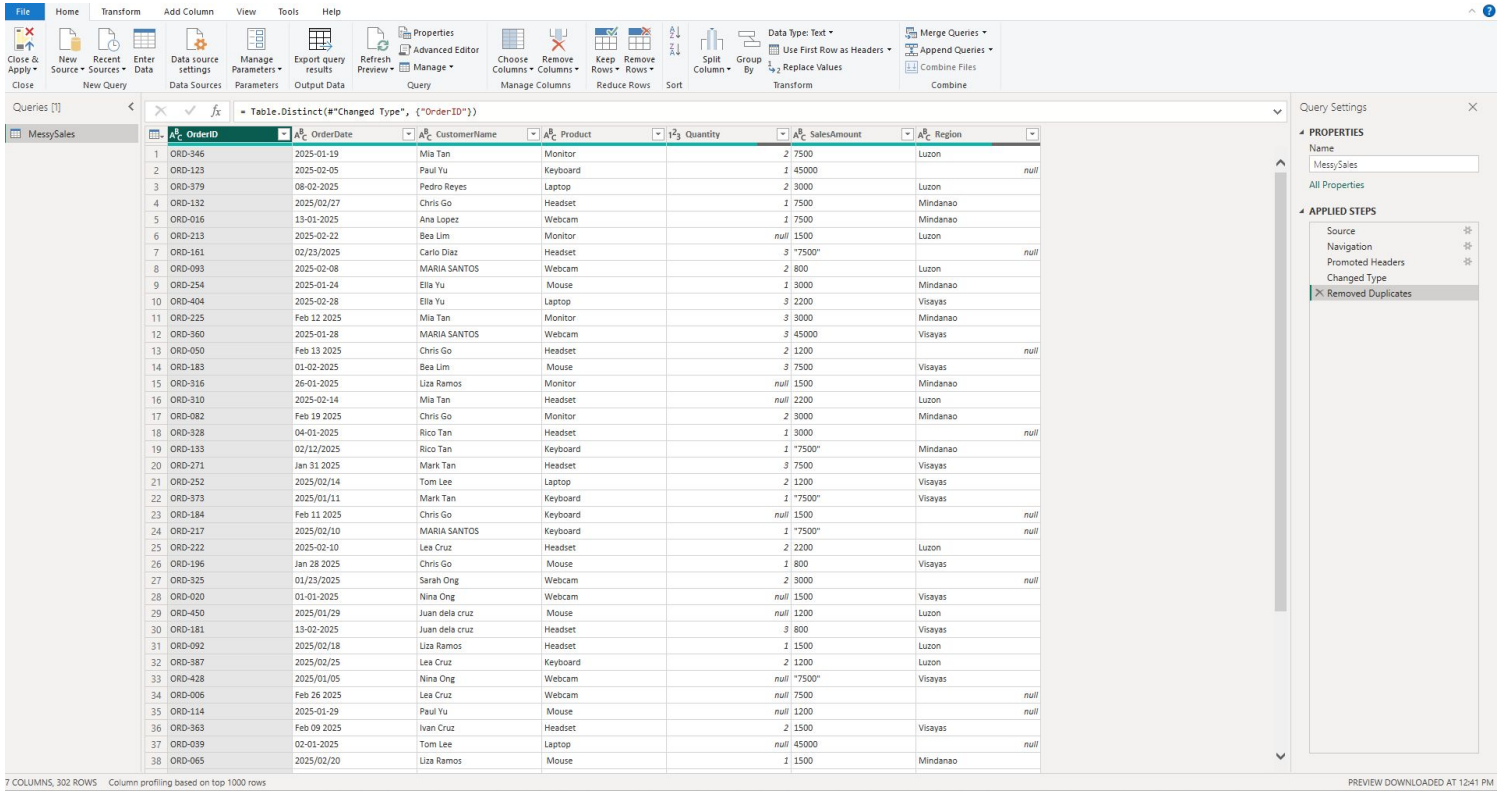
Power Query – Data Cleaning | Part 1: Open Power Query Editor



	OrderID	OrderDate	CustomerName	Product	Quantity	SalesAmount	Region
1	ORD-346	2025-01-19	Mia Tan	Monitor	2	7500	Luzon
2	ORD-123	2025-02-05	Paul Yu	Keyboard	1	45000	null
3	ORD-379	08-02-2025	Pedro Reyes	Laptop	2	3000	Luzon
4	ORD-132	2025/02/27	Chris Go	Headset	1	7500	Mindanao
5	ORD-016	13-01-2025	Ana Lopez	Webcam	1	7500	Mindanao
6	ORD-213	2025-02-22	Bea Lim	Monitor	null	1500	Luzon
7	ORD-161	02/23/2025	Carlo Diaz	Headset	3	"7500"	null
8	ORD-093	2025-02-08	MARIA SANTOS	Webcam	2	800	Luzon
9	ORD-254	2025-01-24	Ella Yu	Mouse	1	3000	Mindanao
10	ORD-404	2025-02-28	Ella Yu	Laptop	3	2200	Visayas
11	ORD-225	Feb 12 2025	Mia Tan	Monitor	3	3000	Mindanao
12	ORD-360	2025-01-28	MARIA SANTOS	Webcam	3	45000	Visayas
13	ORD-050	Feb 13 2025	Chris Go	Headset	2	1200	null
14	ORD-183	01-02-2025	Bea Lim	Mouse	3	7500	Visayas
15	ORD-316	26-01-2025	Liza Ramos	Monitor	null	1500	Mindanao
16	ORD-310	2025-02-14	Mia Tan	Headset	null	2200	Luzon
17	ORD-082	Feb 19 2025	Chris Go	Monitor	2	3000	Mindanao
18	ORD-404	07-02-2025	Sarah Ong	Laptop	3	1200	null
19	ORD-328	04-01-2025	Rico Tan	Headset	1	3000	null
20	ORD-133	02/12/2025	Rico Tan	Keyboard	1	"7500"	Mindanao
21	ORD-271	Jan 31 2025	Mark Tan	Headset	3	7500	Visayas
22	ORD-252	2025/02/14	Tom Lee	Laptop	2	1200	Visayas
23	ORD-373	2025/01/11	Mark Tan	Keyboard	1	"7500"	Visayas
24	ORD-184	Feb 11 2025	Chris Go	Keyboard	null	1500	null
25	ORD-217	2025/02/10	MARIA SANTOS	Keyboard	1	"7500"	null
26	ORD-222	2025-02-10	Lea Cruz	Headset	2	2200	Luzon
27	ORD-316	14-01-2025	Pedro Reyes	Laptop	1	800	Luzon
28	ORD-196	Jan 28 2025	Chris Go	Mouse	1	800	Visayas
29	ORD-325	01/23/2025	Sarah Ong	Webcam	2	3000	null
30	ORD-020	01-01-2025	Nina Ong	Webcam	null	1500	Visayas
31	ORD-450	2025/01/29	Juan dela cruz	Mouse	null	1200	Luzon
32	ORD-181	13-02-2025	Juan dela cruz	Headset	3	800	Visayas
33	ORD-092	2025/02/18	Liza Ramos	Headset	1	1500	Luzon
34	ORD-387	2025/02/25	Lea Cruz	Keyboard	2	1200	Luzon
35	ORD-428	2025/01/05	Nina Ong	Webcam	null	"7500"	Visayas
36	ORD-092	Feb 07 2025	John Lim	Headset	1	"7500"	Mindanao
37	ORD-006	Feb 26 2025	Lea Cruz	Webcam	null	7500	null
38	ORD-114	2025-01-29	Paul Yu	Mouse	null	1200	null

Power Query Editor with the transform data from excel file

Power Query – Data Cleaning | Part 2: Remove Duplicates



	OrderID	OrderDate	CustomerName	Product	Quantity	SalesAmount	Region
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4	ORD-132	2025/02/27	Chris Go	Headset	1	7500	Mindanao
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7	ORD-161	02/23/2025	Carlo Diaz	Headset	3	"7500"	null
8	ORD-093	2025-02-08	MARIA SANTOS	Webcam	2	800	Luzon
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12	ORD-360	2025-01-28	MARIA SANTOS	Webcam	3	45000	Visayas
13	ORD-050	Feb 13 2025	Chris Go	Headset	2	1200	null
14	ORD-183	01-02-2025	Bea Lim	Mouse	3	7500	Visayas
15	ORD-316	26-01-2025	Liza Ramos	Monitor	null	1500	Mindanao
16	ORD-310	2025-02-14	Mia Tan	Headset	null	2200	Luzon
17	ORD-082	Feb 19 2025	Chris Go	Monitor	2	3000	Mindanao
18	ORD-328	04-01-2025	Rico Tan	Headset	1	3000	null
19	ORD-133	02/12/2025	Rico Tan	Keyboard	1	"7500"	Mindanao
20	ORD-271	Jan 31 2025	Mark Tan	Headset	3	7500	Visayas
21	ORD-252	2025/02/14	Tom Lee	Laptop	2	1200	Visayas
22	ORD-373	2025/01/11	Mark Tan	Keyboard	1	"7500"	Visayas
23	ORD-184	Feb 11 2025	Chris Go	Keyboard	null	1500	null
24	ORD-217	2025/02/10	MARIA SANTOS	Keyboard	1	"7500"	null
25	ORD-222	2025-02-10	Lea Cruz	Headset	2	2200	Luzon
26	ORD-196	Jan 28 2025	Chris Go	Mouse	1	800	Visayas
27	ORD-325	01/23/2025	Sarah Ong	Webcam	2	3000	null
28	ORD-020	01-01-2025	Nina Ong	Webcam	null	1500	Visayas
29	ORD-450	2025/01/29	Juan dela cruz	Mouse	null	1200	Luzon
30	ORD-181	13-02-2025	Juan dela cruz	Headset	3	800	Visayas
31	ORD-092	2025/02/18	Liza Ramos	Headset	1	1500	Luzon
32	ORD-387	2025/02/25	Lea Cruz	Keyboard	2	1200	Luzon
33	ORD-428	2025/01/05	Nina Ong	Webcam	null	"7500"	Visayas
34	ORD-006	Feb 26 2025	Lea Cruz	Webcam	null	7500	null
35	ORD-114	2025-01-29	Paul Yu	Mouse	null	1200	null
36	ORD-363	Feb 09 2025	Ivan Cruz	Headset	2	1500	Visayas
37	ORD-039	02-01-2025	Tom Lee	Laptop	null	45000	null
38	ORD-065	2025/02/20	Liza Ramos	Mouse	1	1500	Mindanao

OrderID column got to be unique so duplicates were removed

Power Query – Data Cleaning | Part 3: Handle Missing Values

Untitled - Power Query Editor

File | Home | Transform | Add Column | View | Tools | Help

Table: Group By, Use First Row as Headers, Transpose, Reverse Rows, Count Rows
Any Column: Data Type: Text, Detect Data Type, Rename
Text Column: Split Column, Format, Merge Columns, Extract, Parse
Number Column: Statistics, Standard, Scientific, Trigonometry, Rounding, Information
Date & Time Column: Date, Time, Duration
Scripts: Run R script, Run Python script

Queries [1]

MessySales

```
= Table.FillDown(#"Removed Blank Rows1",{"Quantity", "Region"})
```

	Product	Quantity	SalesAmount	Region
1	Monitor	2	7500	Luzon
2	Keyboard	1	45000	0
3	Laptop	2	3000	Luzon
4	Headset	1	7500	Mindanao
5	Webcam	1	7500	Mindanao
6	Monitor	0	1500	Luzon
7	Headset	3	"7500"	0
8	Webcam	2	800	Luzon
9	Mouse	1	3000	Mindanao
10	Laptop	3	2200	Visayas
11	Monitor	3	3000	Mindanao
12	Webcam	3	45000	Visayas
13	Headset	2	1200	0
14	Mouse	3	7500	Visayas
15	Monitor	0	1500	Mindanao
16				

Query Settings

PROPERTIES

Name: MessySales

All Properties

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Removed Duplicates
- Replaced Value
- Replaced Value1
- Removed Blank Rows
- Removed Blank Rows
- Filled Down

7 COLUMNS, 302 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 5:45 PM

Handling missing value by replacing values and removing blank rows

Power Query – Data Cleaning | Part 4: Change Data Types

Untitled - Power Query Editor

File | Home | Transform | Add Column | View | Tools | Help

Group By | Use First Row as Headers | Transpose | Reverse Rows | Count Rows | Table

Data Type: Decimal Number | Detect Data Type | Rename | Any Column

Split Column | Format | Merge Columns | Extract | Parse | Text Column

Statistics | Standard | Scientific | Trigonometry | Rounding | Information | Number Column

Date | Time | Duration | Date & Time Column

Run R script | Run Python script | Scripts

Queries

```
= Table.TransformColumnTypes(#"Filled Down",{{"OrderDate", type date}, {"SalesAmount", type number},
```

	OrderDate	CustomerName	Product	Quantity	SalesAmount
1	1/19/2025	Mia Tan	Monitor	2	7500
2	2/5/2025	Paul Yu	Keyboard	1	45000
3	8/2/2025	Pedro Reyes	Laptop	2	3000
4	2/27/2025	Chris Go	Headset	1	7500
5	ror	Ana Lopez	Webcam	1	7500
6	2/22/2025	Bea Lim	Monitor	0	1500
7	2/23/2025	Carlo Diaz	Headset	3	Error
8	2/8/2025	MARIA SANTOS	Webcam	2	800
9	1/24/2025	Ella Yu	Mouse	1	3000
10	2/28/2025	Ella Yu	Laptop	3	2200
11	2/12/2025	Mia Tan	Monitor	3	3000
12	1/28/2025	MARIA SANTOS	Webcam	3	45000
13	2/13/2025	Chris Go	Headset	2	1200
14	1/2/2025	Bea Lim	Mouse	3	7500
15	ror	Liza Ramos	Monitor	0	1500
16					

Query Settings

PROPERTIES

Name: MessySales

All Properties

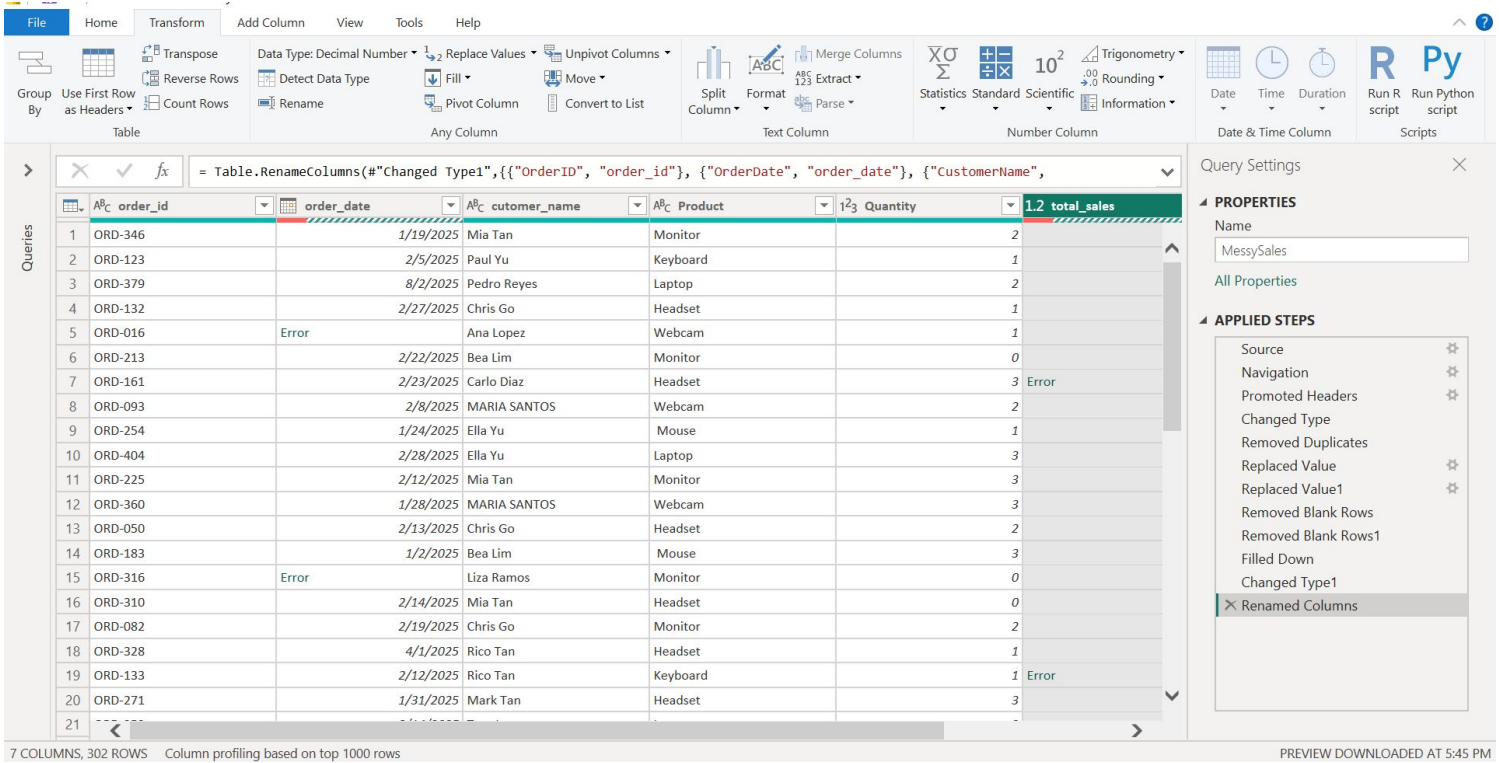
APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Removed Duplicates
- Replaced Value
- Replaced Value1
- Removed Blank Rows
- Removed Blank Rows1
- Filled Down
- Changed Type1

7 COLUMNS, 302 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 5:45 PM

Changing the data type of each value in accordance to its type

Power Query – Data Cleaning | Part 5: Rename Columns Properly



Renaming columns into standard names

Power Query – Data Cleaning | Guide Questions

1. Why is removing duplicates important in data analysis?

Removing duplicates is important because duplicate records can distort totals, inflate calculations, and lead to inaccurate analysis and poor decision-making.

2. What problems can missing values cause in reports?

Missing values can cause incorrect totals and averages, errors in calculations, and incomplete or misleading reports and visualizations.

3. How does correct data typing affect calculations?

Correct data typing ensures accurate calculations, proper sorting and filtering, and correct date-based analysis, preventing errors in reports.

4. What naming conventions improve data readability?

Clear and consistent naming conventions, such as using lowercase letters, underscores, and descriptive names like `order_id` and `total_sales`, improve readability, organization, and data usability.

Power Query – Data Cleaning | Enhancement Activities

	OrderID	OrderDate	CustomerName
980	null	null	null
981	null	null	null
982	null	null	null
983	null	null	null
984	null	null	null
985	null	null	null
986	null	null	null
987	null	null	null
988	null	null	null
989	null	null	null
990	null	null	null
991	null	null	null
992	null	null	null
993	null	null	null
994	null	null	null
995	null	null	null
996	null	null	null
997	null	null	null
998	null	null	null
999	null	null	null

Before cleaning the data

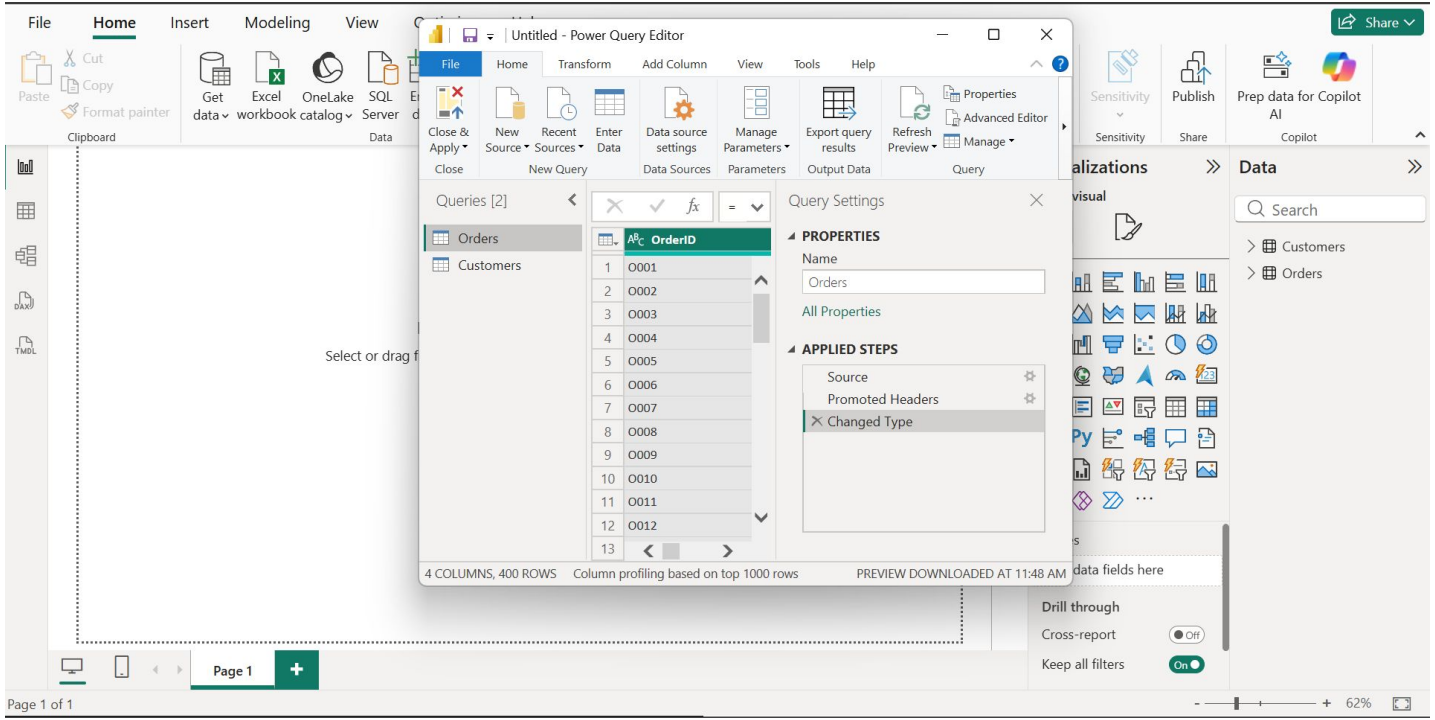
	order_id	order_date	customer_n
190	ORD-391	2/10/2025	Liza Ramos
191	ORD-342	1/13/2025	Ana Lopez
192	ORD-242	2/1/2025	Carlo Diaz
193	ORD-419	7/2/2025	Mark Tan
194	ORD-368	3/2/2025	Ella Yu
195	ORD-246	1/24/2025	Ivan Cruz
196	ORD-056	2/10/2025	Chris Go
197	ORD-333	1/28/2025	Tom Lee
198	ORD-303	1/19/2025	Nina Ong
199	ORD-416	1/4/2025	Mark Tan
200	ORD-146	2/8/2025	Ana Lopez
201	ORD-104	1/12/2025	Rico Tan
202	ORD-108	2/1/2025	Pedro Reyes
203	ORD-327	2/5/2025	Nina Ong
204	ORD-229	1/3/2025	Mark Tan
205	ORD-331	1/14/2025	Paul Yu
206	ORD-172	1/27/2025	Mia Tan
207	ORD-021	1/24/2025	John Lim
208	ORD-238	2/16/2025	Paul Yu
209	ORD-256	6/1/2025	Pedro Reyes

After cleaning the data

Research: What is ETL and how does Power Query support it?

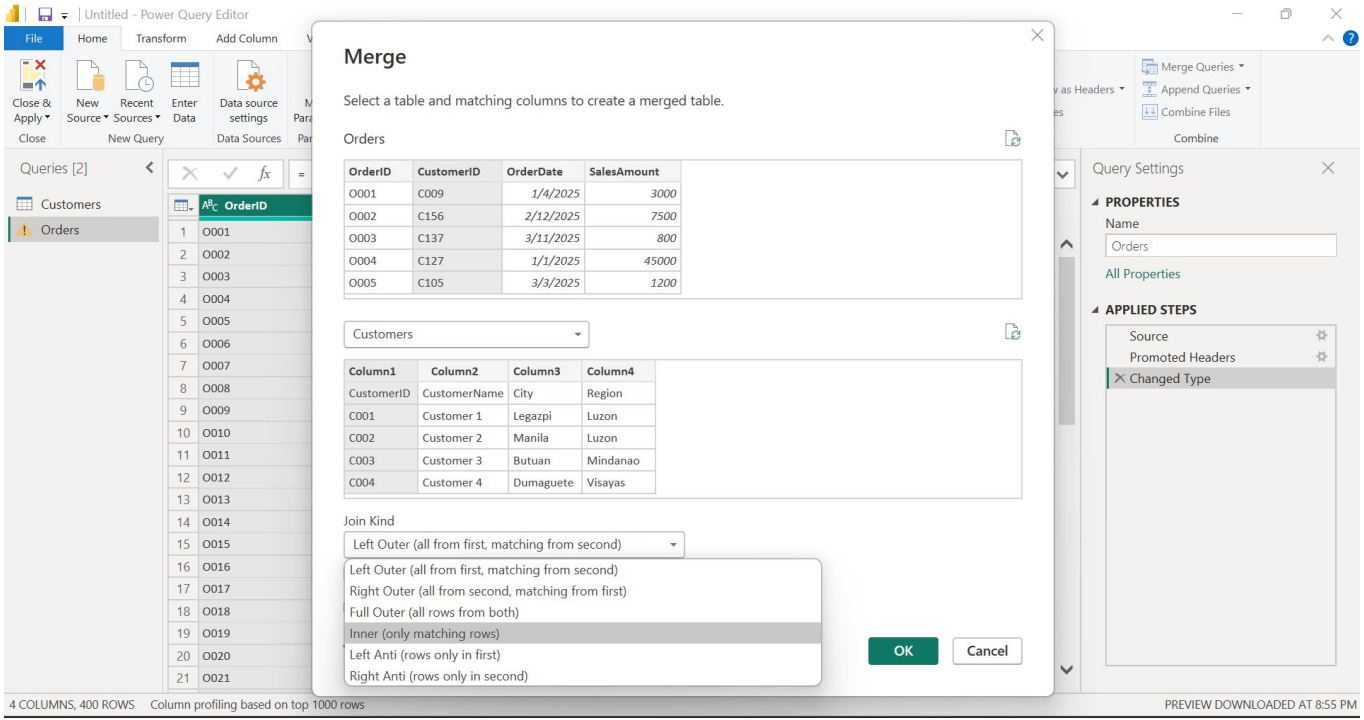
ETL stands for **Extract, Transform, and Load**, which is a process used in data analytics to collect data from different sources, clean or modify it, and load it into a system for analysis. Power Query supports ETL by allowing users to **extract data from sources like Excel or databases, transform it through cleaning operations such as removing duplicates and changing data types, and load the processed data into Power BI for reporting and visualization.**

Merge and Append Queries | Part 1: Load Data



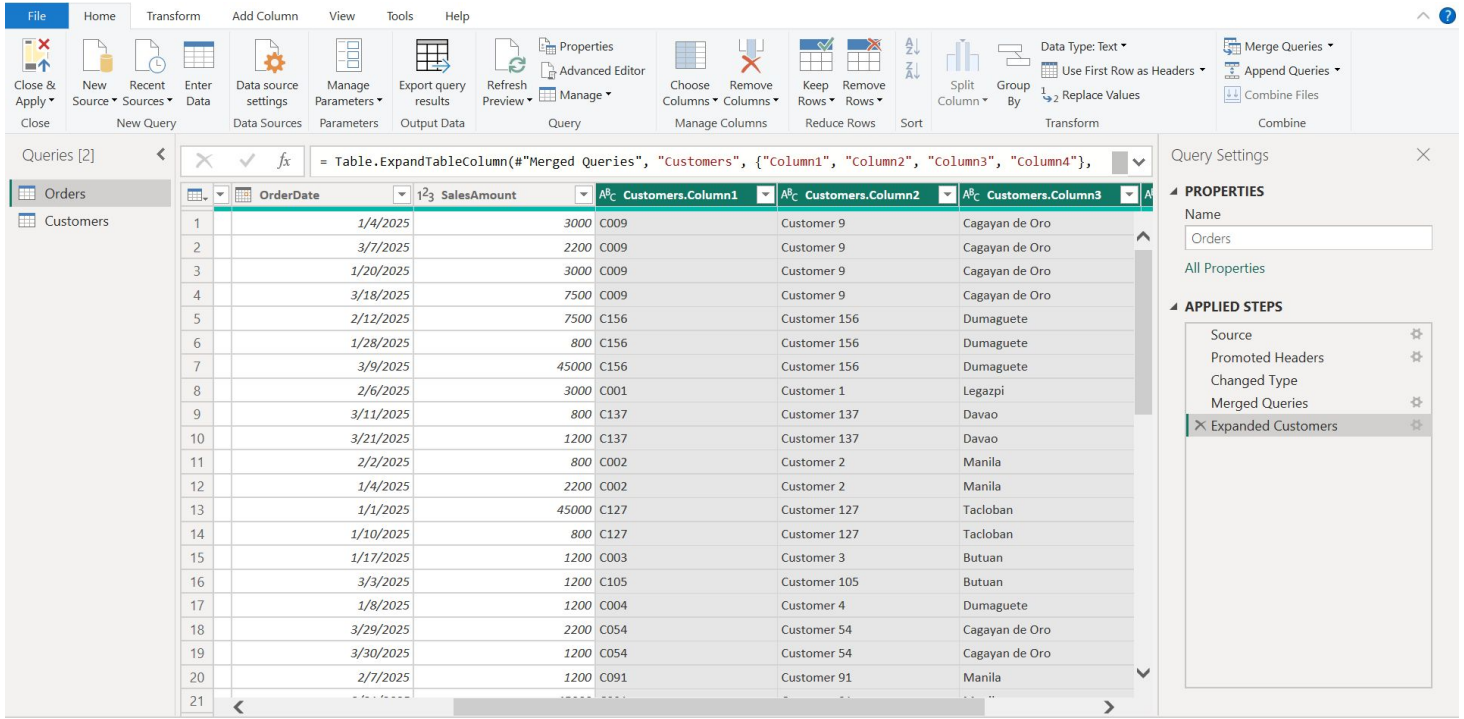
Load two different data table and opened query editor.

Merge and Append Queries | Part 2: Merge Datasets (Inner Join)



Inner Joining two tables using the CustomerID as common value

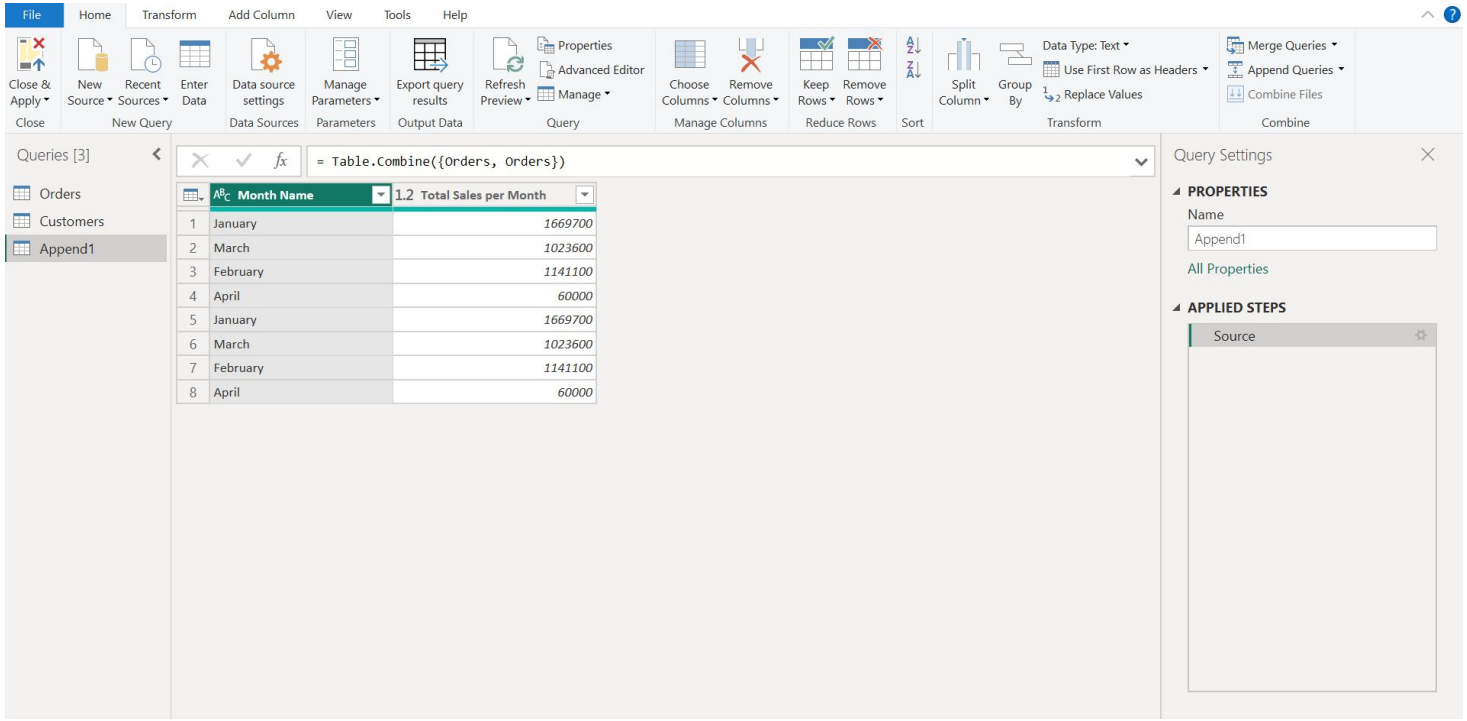
Merge and Append Queries | Part 2: Merge Datasets (Inner Join)



	OrderDate	SalesAmount	Customers.Column1	Customers.Column2	Customers.Column3
1	1/4/2025	3000	C009	Customer 9	Cagayan de Oro
2	3/7/2025	2200	C009	Customer 9	Cagayan de Oro
3	1/20/2025	3000	C009	Customer 9	Cagayan de Oro
4	3/18/2025	7500	C009	Customer 9	Cagayan de Oro
5	2/12/2025	7500	C156	Customer 156	Dumaguete
6	1/28/2025	800	C156	Customer 156	Dumaguete
7	3/9/2025	45000	C156	Customer 156	Dumaguete
8	2/6/2025	3000	C001	Customer 1	Legazpi
9	3/11/2025	800	C137	Customer 137	Davao
10	3/21/2025	1200	C137	Customer 137	Davao
11	2/2/2025	800	C002	Customer 2	Manila
12	1/4/2025	2200	C002	Customer 2	Manila
13	1/1/2025	45000	C127	Customer 127	Tacloban
14	1/10/2025	800	C127	Customer 127	Tacloban
15	1/17/2025	1200	C003	Customer 3	Butuan
16	3/3/2025	1200	C105	Customer 105	Butuan
17	1/8/2025	1200	C004	Customer 4	Dumaguete
18	3/29/2025	2200	C054	Customer 54	Cagayan de Oro
19	3/30/2025	1200	C054	Customer 54	Cagayan de Oro
20	2/7/2025	1200	C091	Customer 91	Manila
21					

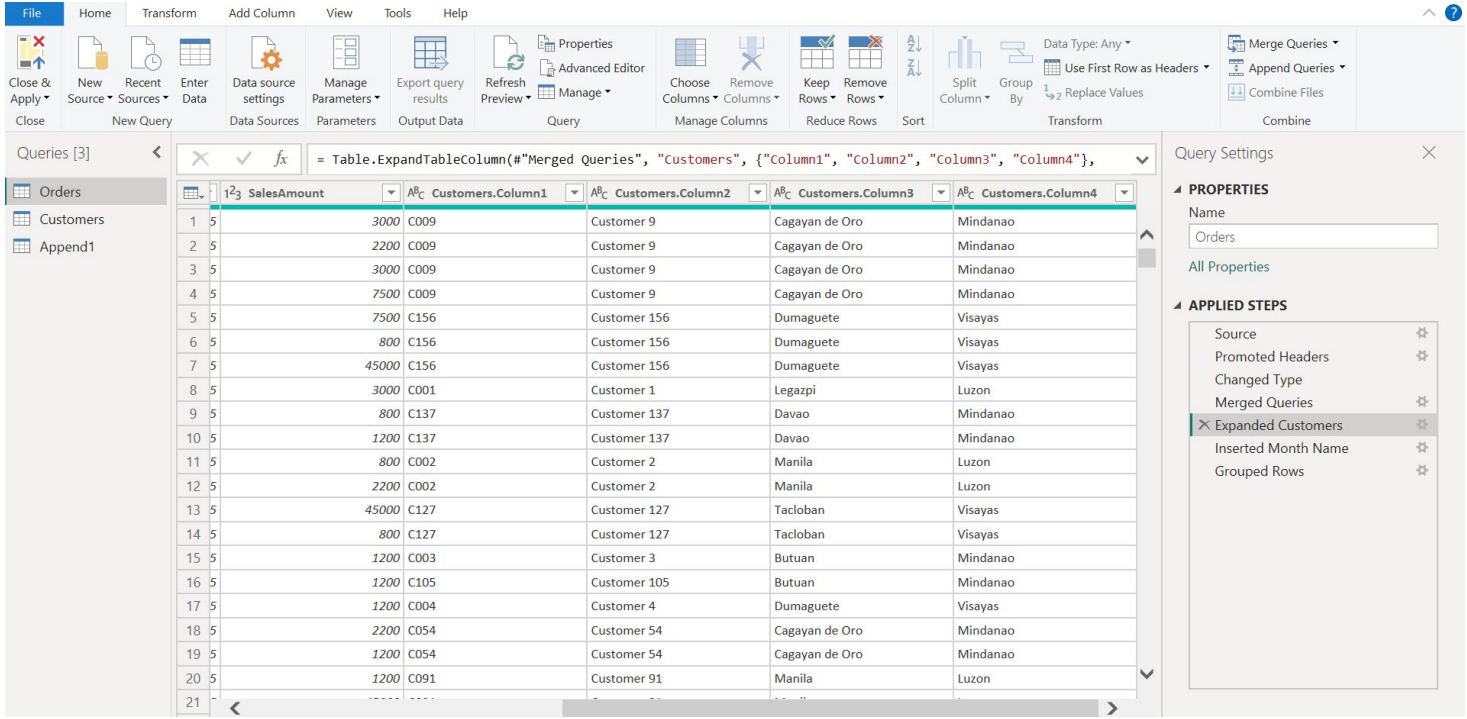
Inner Joining two tables and expanding the table

Merge and Append Queries | Part 3: Append Monthly Sales Files



Append new to separate the multiple monthly sales

Merge and Append Queries | Part 4: Validate Merged Data



		SalesAmount	Customers.Column1	Customers.Column2	Customers.Column3	Customers.Column4
1	5	3000	C009	Customer 9	Cagayan de Oro	Mindanao
2	5	2200	C009	Customer 9	Cagayan de Oro	Mindanao
3	5	3000	C009	Customer 9	Cagayan de Oro	Mindanao
4	5	7500	C009	Customer 9	Cagayan de Oro	Mindanao
5	5	7500	C156	Customer 156	Dumaguete	Visayas
6	5	800	C156	Customer 156	Dumaguete	Visayas
7	5	45000	C156	Customer 156	Dumaguete	Visayas
8	5	3000	C001	Customer 1	Legazpi	Luzon
9	5	800	C137	Customer 137	Davao	Mindanao
10	5	1200	C137	Customer 137	Davao	Mindanao
11	5	800	C002	Customer 2	Manila	Luzon
12	5	2200	C002	Customer 2	Manila	Luzon
13	5	45000	C127	Customer 127	Tacloban	Visayas
14	5	800	C127	Customer 127	Tacloban	Visayas
15	5	1200	C003	Customer 3	Butuan	Mindanao
16	5	1200	C105	Customer 105	Butuan	Mindanao
17	5	1200	C004	Customer 4	Dumaguete	Visayas
18	5	2200	C054	Customer 54	Cagayan de Oro	Mindanao
19	5	1200	C054	Customer 54	Cagayan de Oro	Mindanao
20	5	1200	C091	Customer 91	Manila	Luzon
21						

Confirming merged data

Power Query – Merge and Append Queries | Guide Questions

1. What is the difference between Merge and Append?

Merge combines two tables by matching a common column like CustomerID and adds related columns. Append combines tables by adding rows from one table to another.

2. Why use Inner Join instead of Left Join?

Inner Join is used to keep only records that exist in both tables, ensuring the data matches correctly.

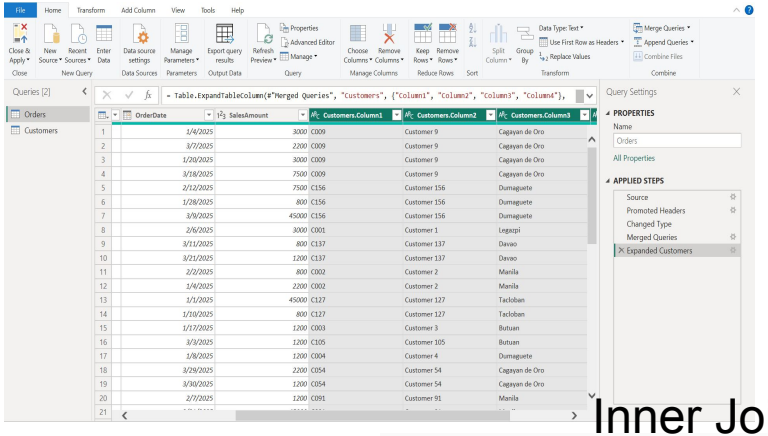
3. What happens to unmatched records in an Inner Join?

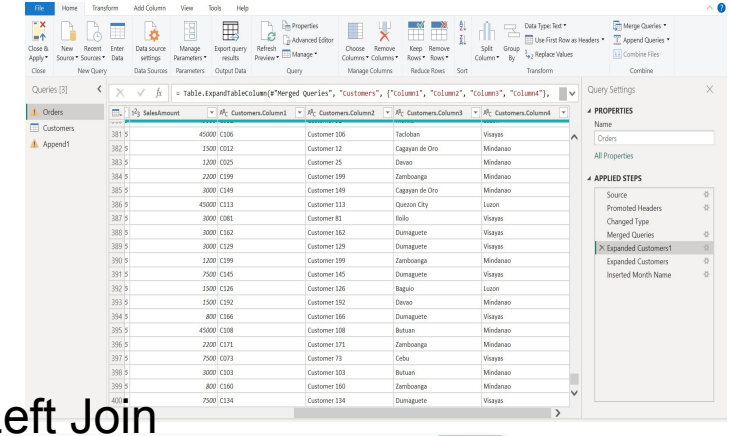
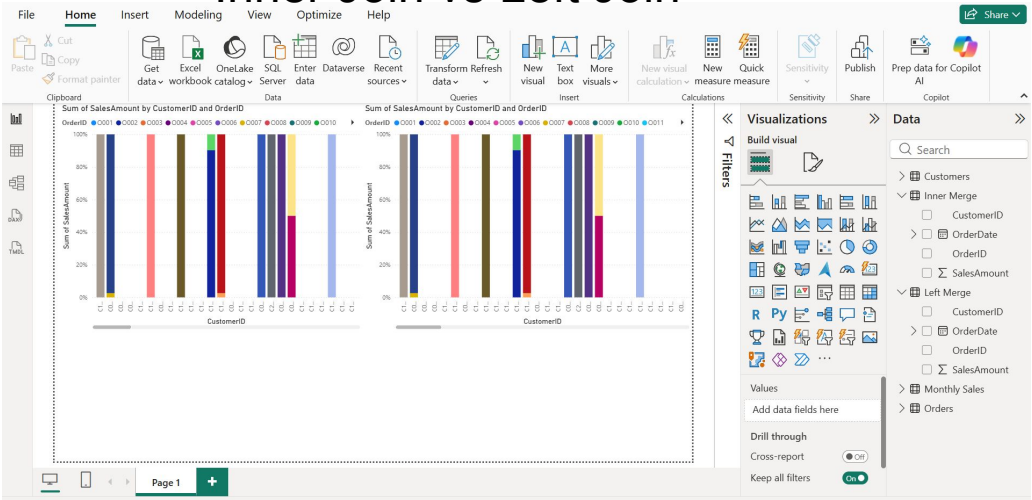
Unmatched records are removed and will not appear in the merged table.

4. When is Append more appropriate than Merge?

Append is used when combining tables with the same columns, such as monthly sales data. Instead of matching data, append simply adds the rows from one table to another to create one larger dataset for analysis.

Merge and Append Queries | Enhancement Activities





Inner Join vs Left Join